

From Data to Models: Analyzing and Integrating Biological Data into Mechanistic Models

Dora Tortarolo, Simone Pernice, Francesca Cordero

Course of *Doctoral school*
PhD in Complex Systems for Quantitative Biomedicine

Who we are



Simone Pernice is currently assistant professor at the Department of Computer Science of the University of Turin, coordinator of Young InfoLife group of the Infolife National Laboratory of CINI. His current research activities mainly involve computational epidemiology, modeling of complex biological systems, and longitudinal data analysis.

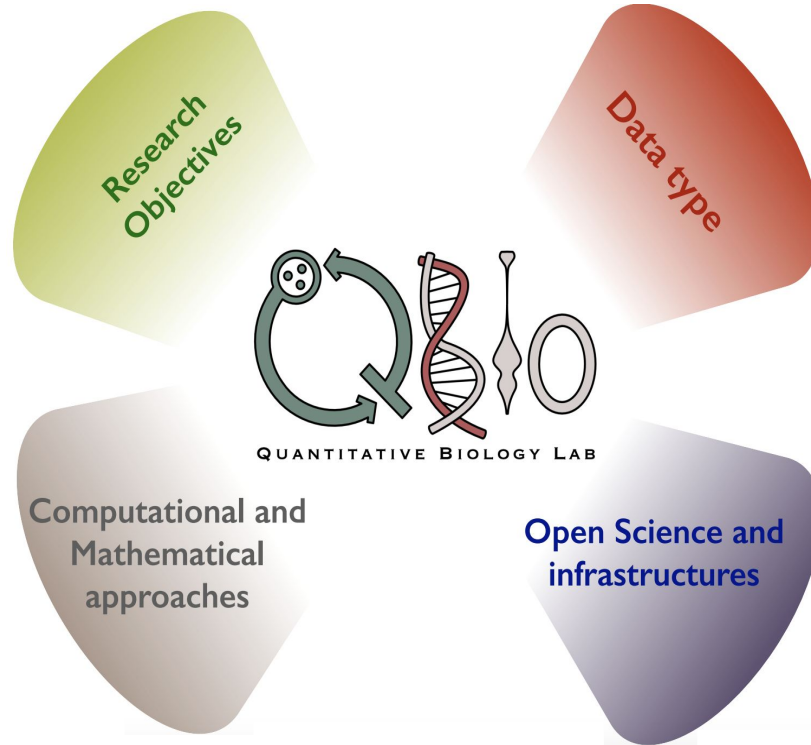
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Dora Tortarolo is currently a Post Doc at the Department of Computer Science of the University of Turin. Biologist by training, she pursued her PhD both at the biology lab bench and in the computer lab, receiving in 2022 her PhD degree in “Complex Systems for the Life Sciences” at the University of Turin. Currently, she focuses on the microbiome data analysis and modeling of complex biological systems.

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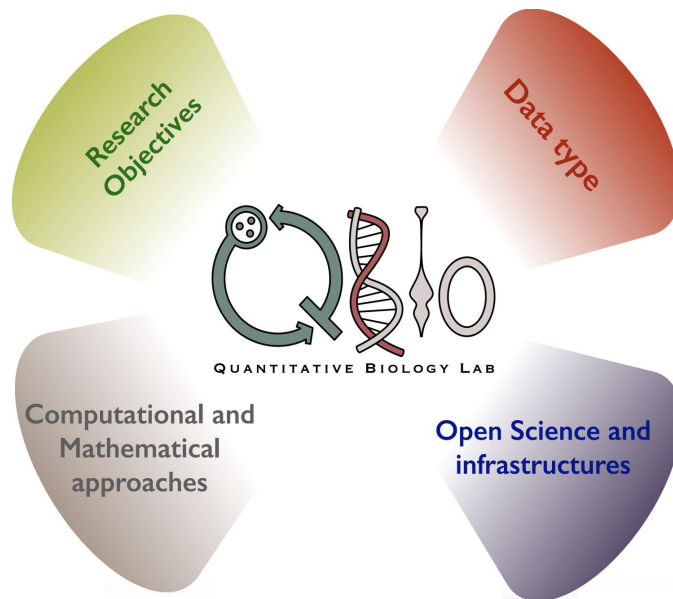
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Research Objectives:

- Patient stratification
- Identification of diagnostic and prognostic biomarkers
- Epidemiological models
- Mathematical models for cancer progression and treatment response
- Evolutionary Genomics and Modelling
- Basic Research: Cancer Genomics and Epigenomics
- Microbiome research: development of data analysis pipelines



When?

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
14	15	16	17	18	19 • 09:00 - 13:00 Sala Seminari 	20
21 • 09:00 - 13:00 Sala conferenze, terzo piano 	22	23	24	25	26 • 09:00 - 13:00 Sala conferenze, terzo piano 	27
28 • 09:00 - 13:00 Sala conferenze, terzo piano 	29	30	1	2	3	4

Course Outline

Day 1 – Introduction to Mechanistic Modeling and the Importance of Data

- Welcome and Course Introduction
- Overview of computational models in biology.
- Mechanistic approaches: conceptual foundations of Petri Nets and the *epimod* framework.
- Hands-on Activity: the Schlögl model to understand system dynamics and sensitivity.

Day 2 – Calibration and Predictive Modeling

- Model Calibration: introduction to calibration techniques for mechanistic models.
- Predictive Simulations: Performing what-if analyses using calibrated models.
- Sensitivity analysis (optional)

Day 3 – From Case Study to Data Analysis

<https://github.com/qBioTurin/From-Data-to-Models.git>

- From Data to Models: the importance of model parameterization using real (raw) experimental data.
- Real-World Case Study: understanding the problem context.
- Real-World Case Study: model definition.
- Real-World Case Study: model calibration.

Day 4 – Data Integration and Advanced Modeling

- Flux Balance Analysis: Principles and applications in systems biology.
- Integrative Modeling: combining FBA with experimental data for enhanced predictions.
- Introduction to UnifiedGreatMOD: linking heterogeneous data to mechanistic and constraint-based models.
- Hands-on: FBA case study with data integration.
- Course Wrap-up: Summary of key concepts.



Group Work

During the course, you will work on the hands-on exercises in small groups of **2–3 people**.



<https://github.com/qBioTurin/From-Data-to-Models.git>

Groups will be **defined during the break**, so that everyone can collaborate and discuss the exercises together.

